

THE GEORGE WASHINGTON UNIVERSITY
WASHINGTON, DC

Mechanical & Aersopace Engineering: The Many Facets of Energy

Saniya LeBlanc

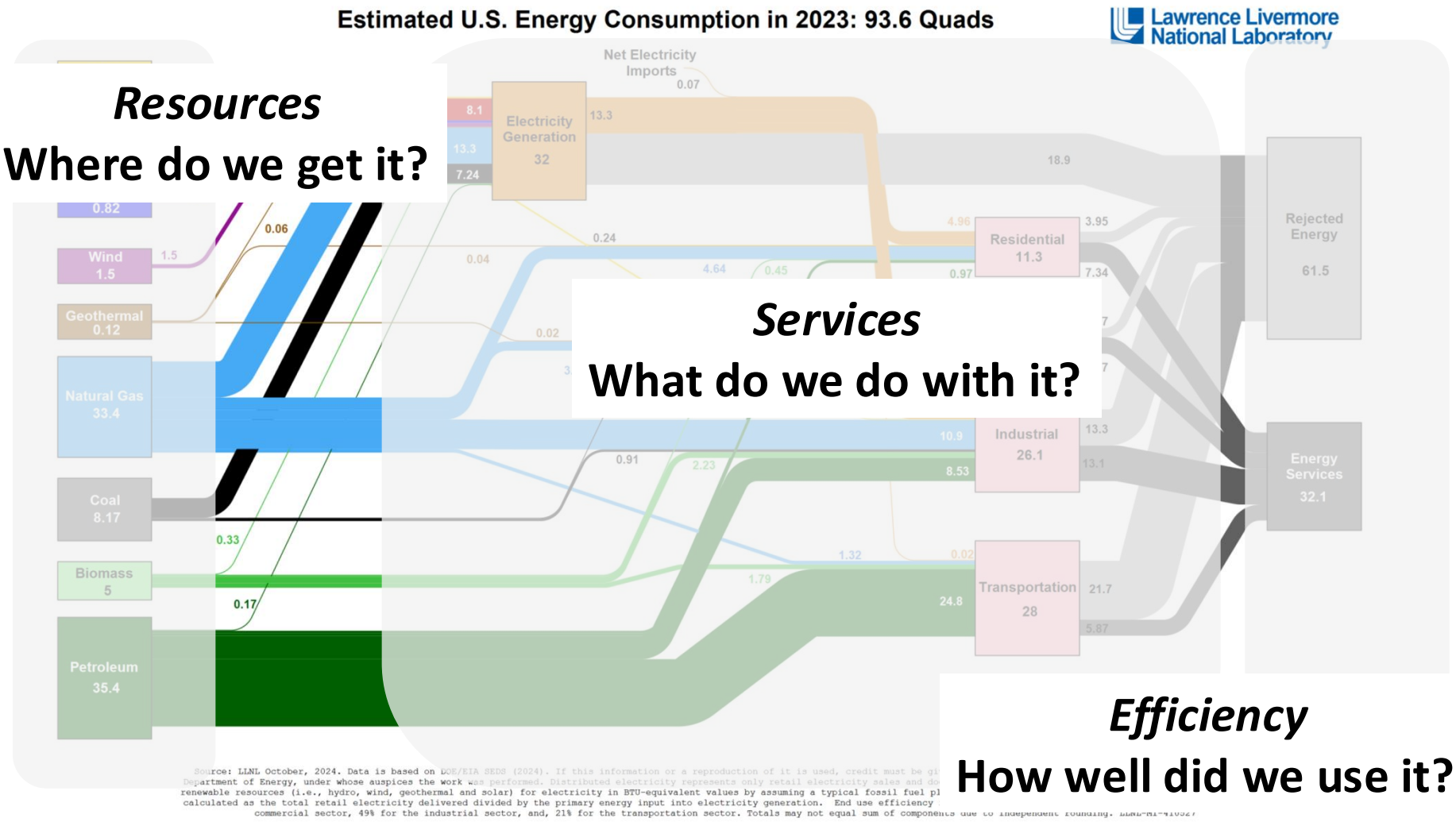
Professor

Mechanical & Aerospace Engineering

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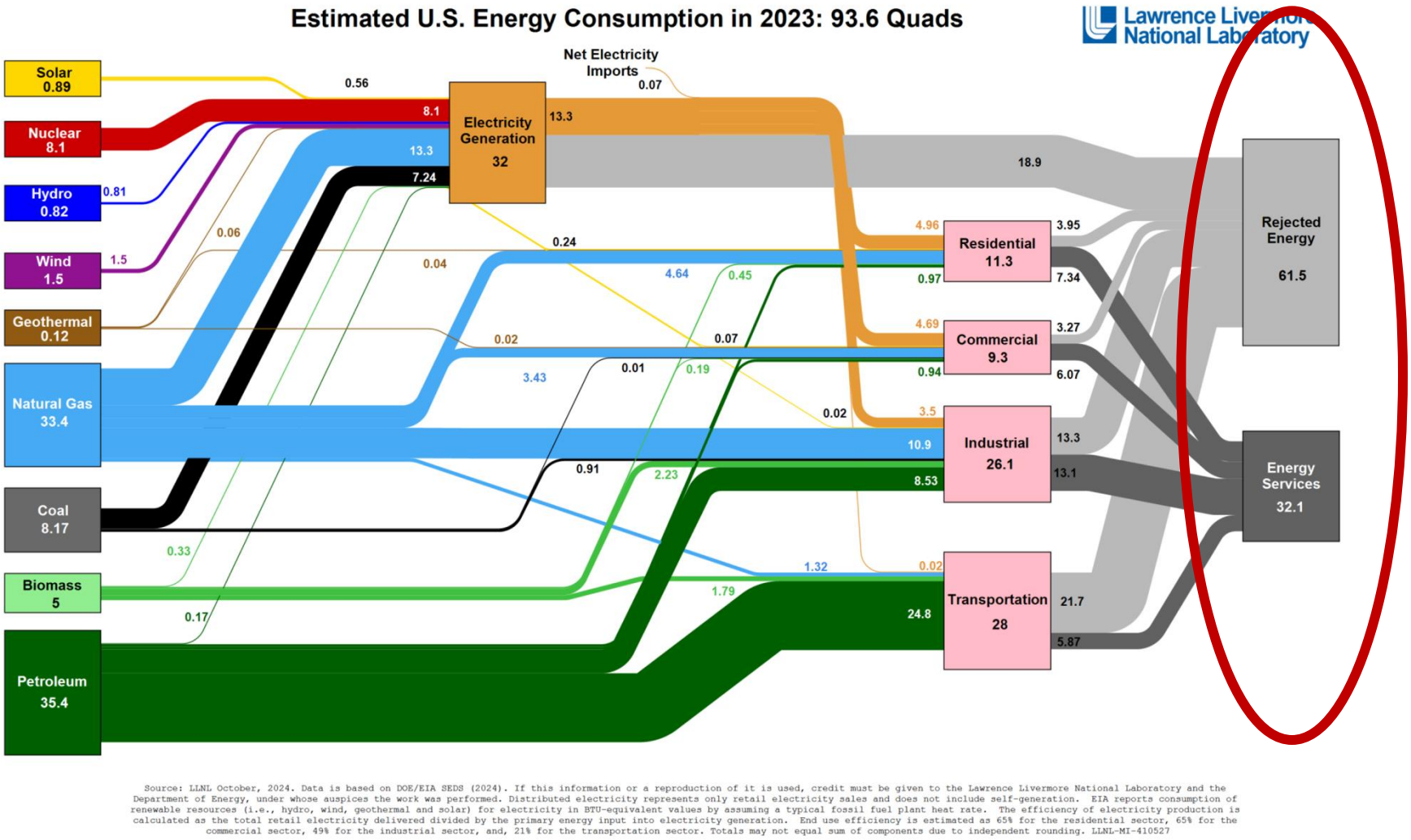
www.leblanclab.com

Energy Flows



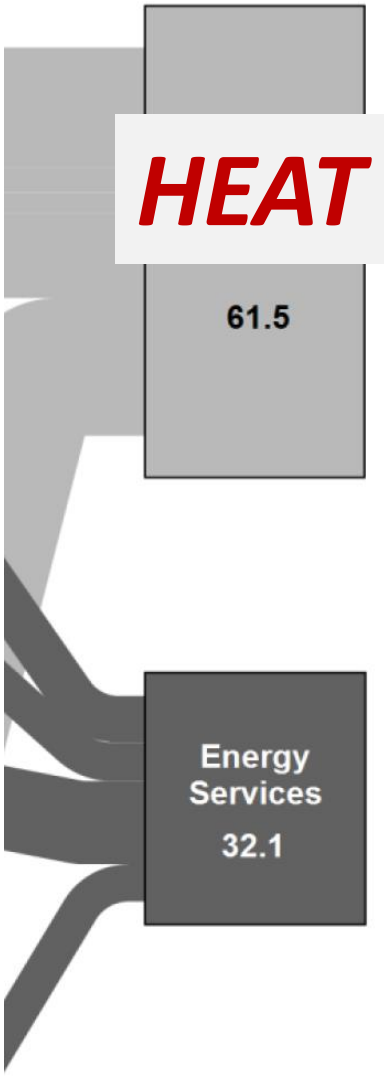
https://flowcharts.llnl.gov/commodities/energy

Where did all that energy go?



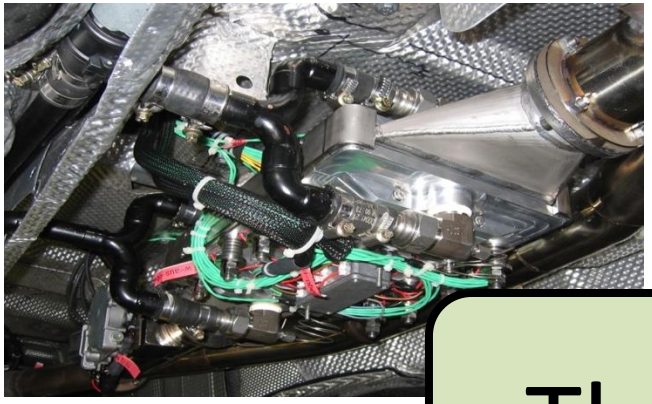
https://flowcharts.llnl.gov/commodities/energy

Where did all that energy go?



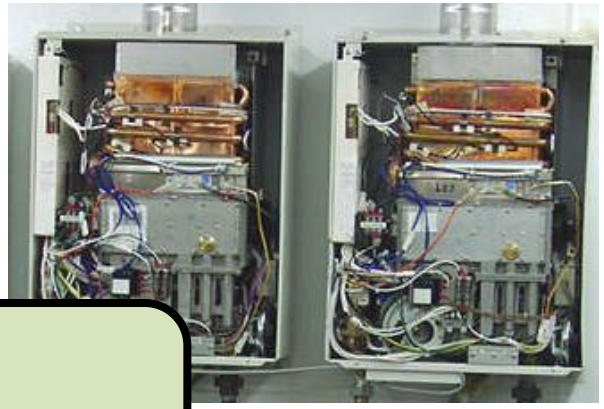
Energy Systems

Automotive Exhaust



www.bosch.com

Water Heater



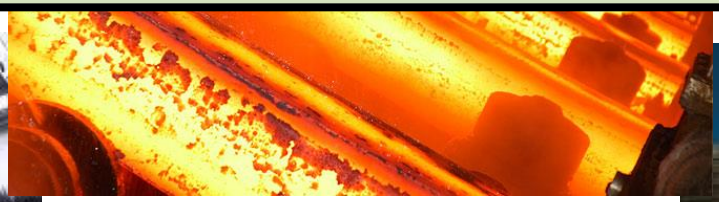
www.water.arredemo.org

Thermal Systems

Gas Turbine



www.usa.siemens.com



www.risenterprises.com

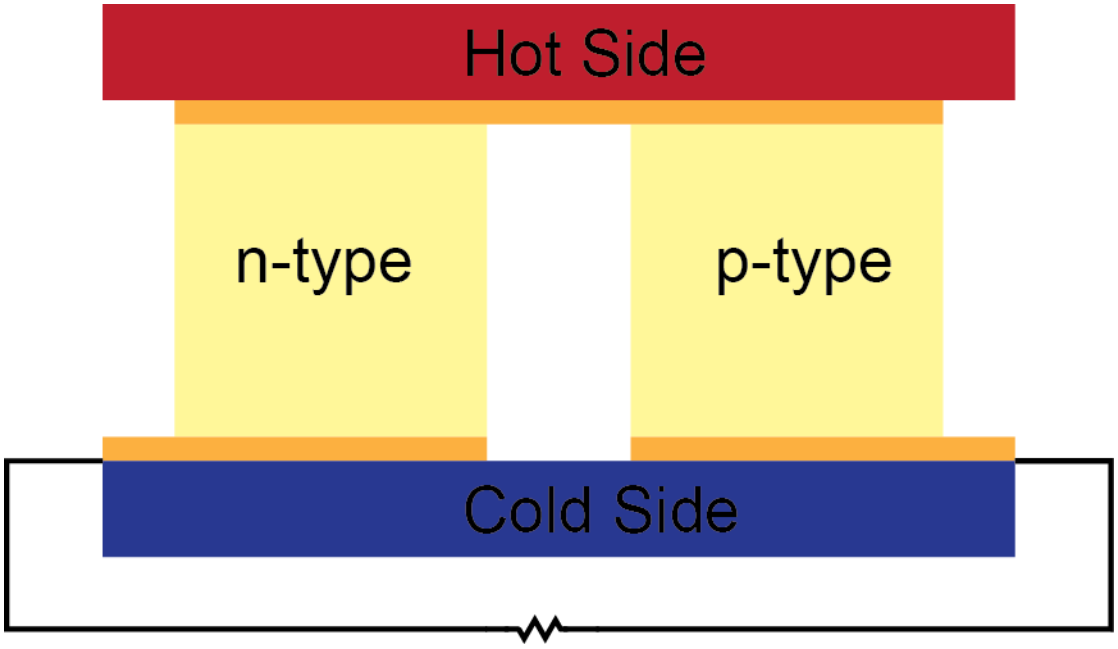
Glass Lehr



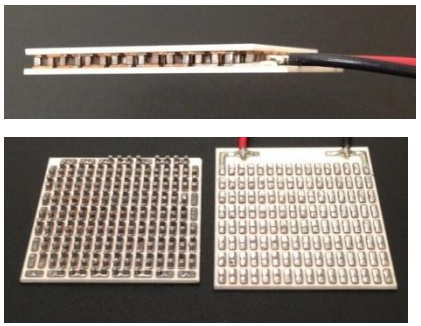
www.glassfurnacemaker.com

Thermoelectric Energy Conversion

Power generation: Convert thermal energy to electrical energy.



Thermoelectric Device



Advantages:

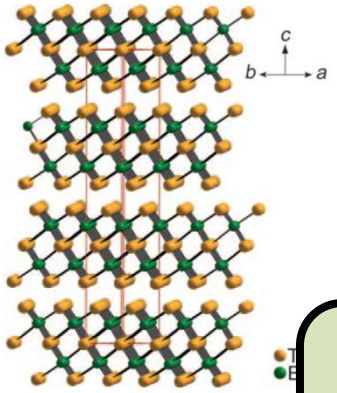
Distributed generation
Localized control of energy transfer
Reliable
No moving parts
Silent

Challenges:

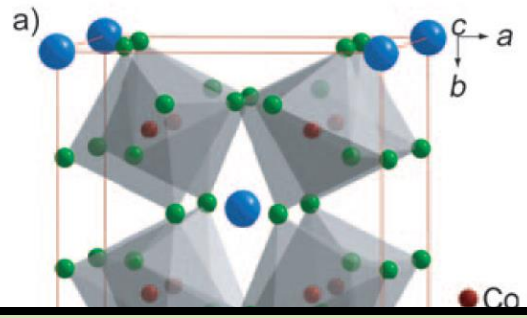
System integration
Conversion efficiency
Operating temperatures

Thermoelectric Materials

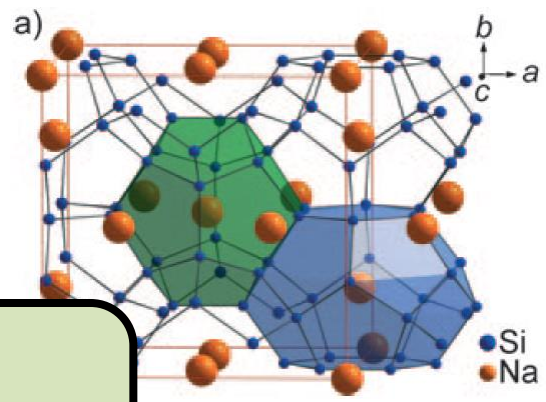
Chalcogenide



Skutterudite

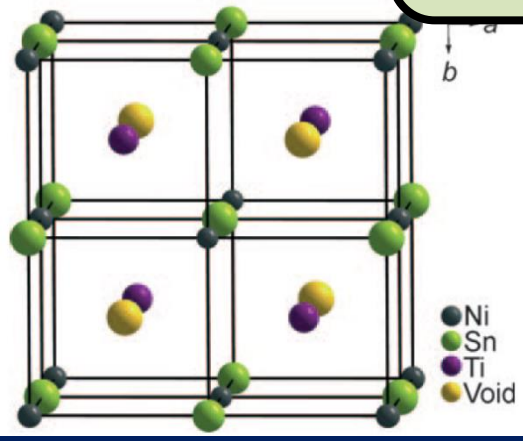


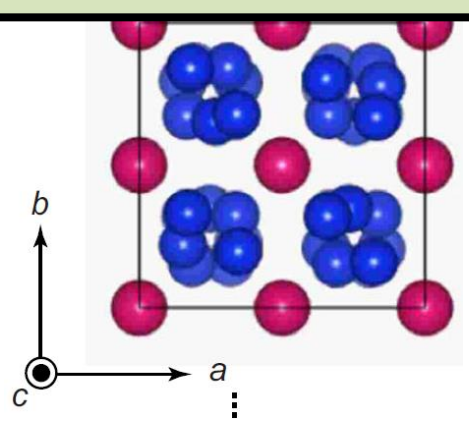
Clathrate



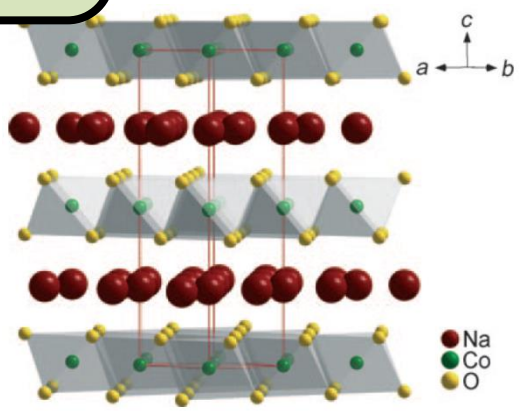
Materials Science

Half Heusler





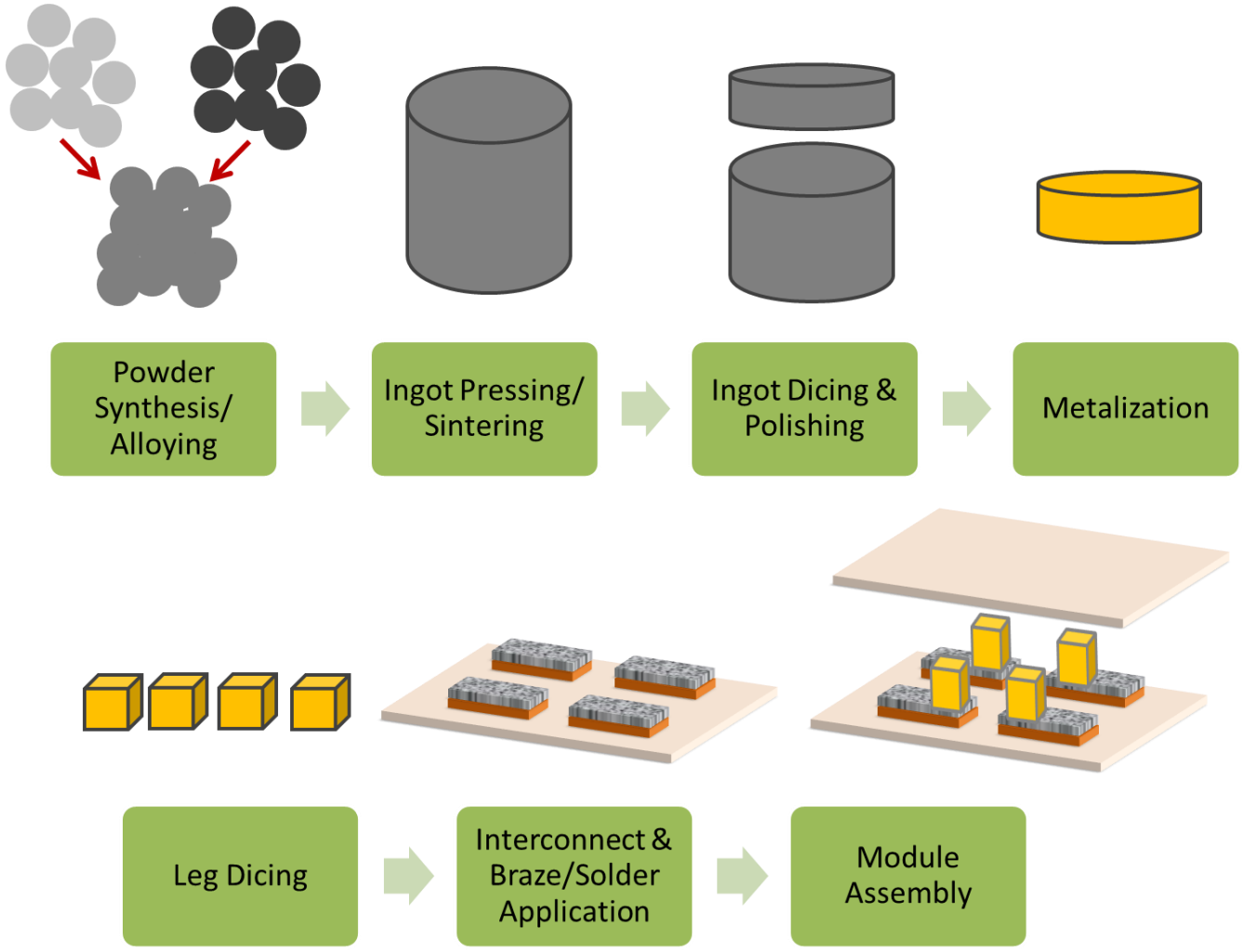
Metal Oxide



Sootsman *et al.*, *Angew .Chem.* (2009)

Miyazaki *et al.*, *Phys. Rev. B* (2008)

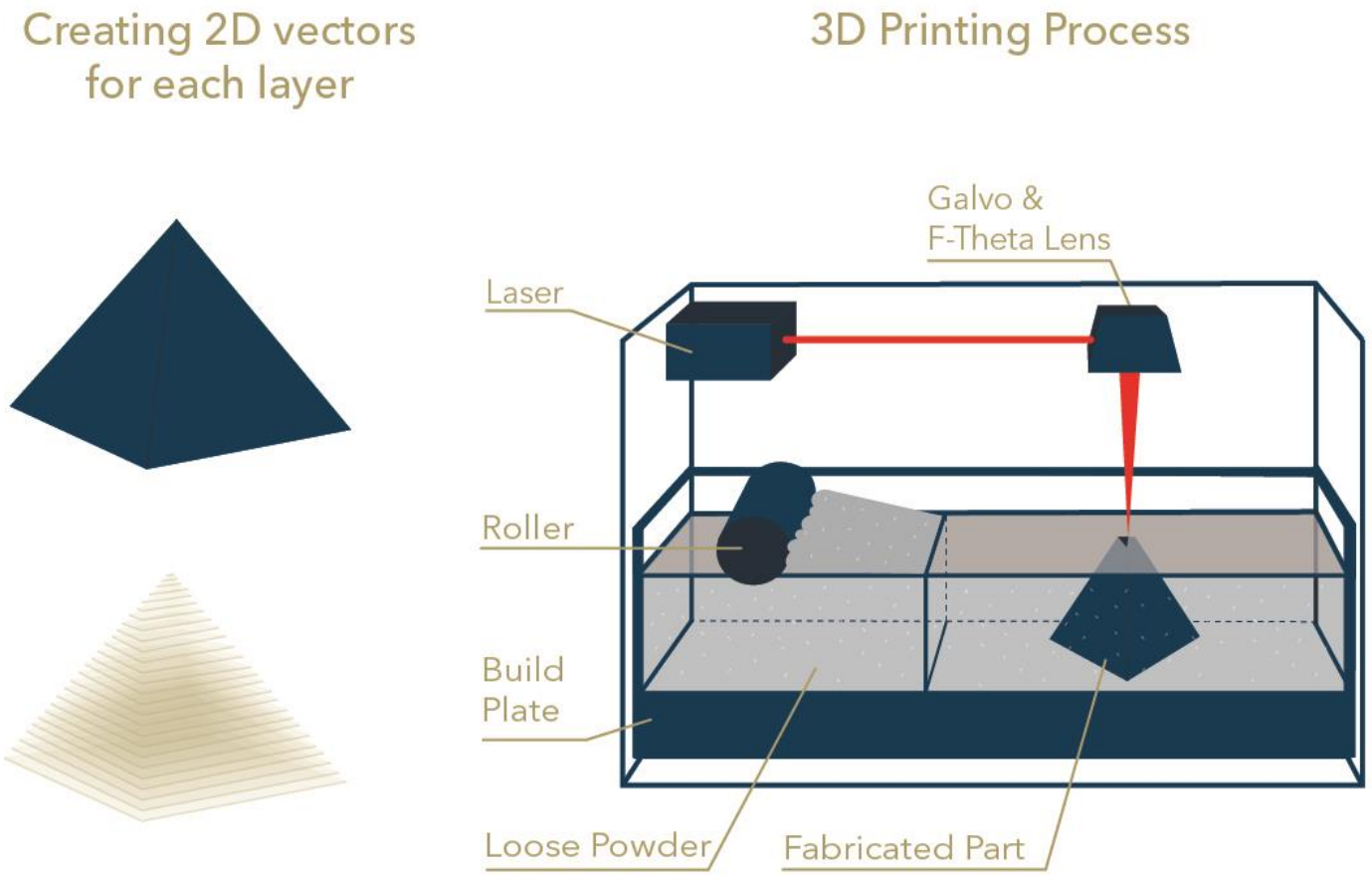
Device Manufacturing



S. LeBlanc, "Thermoelectric Generators: Linking Material Properties and Systems Engineering for Waste Heat Recovery," *Sustainable Materials & Technologies* (2014)

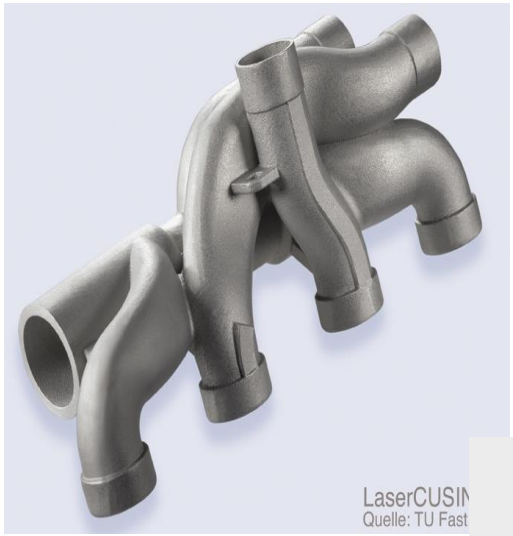
Additive Manufacturing

Laser Powder Bed Fusion:

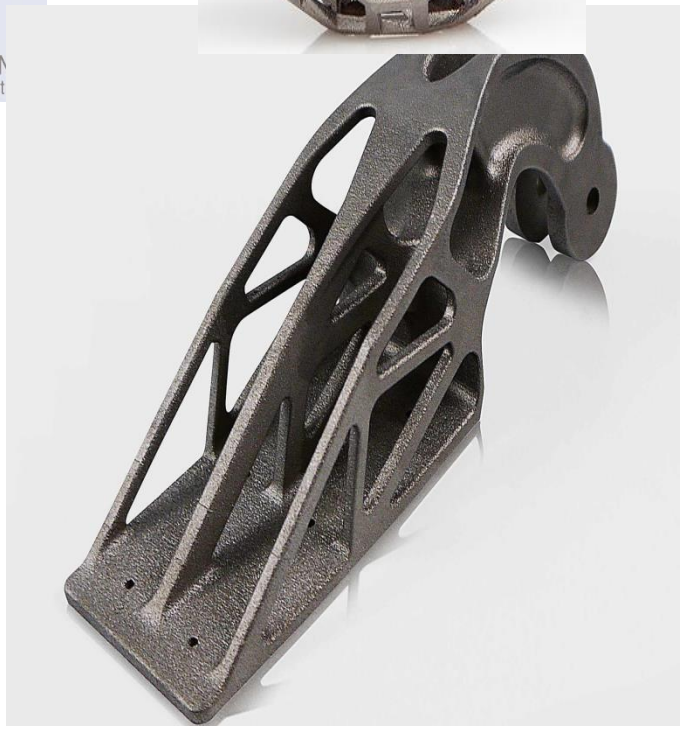


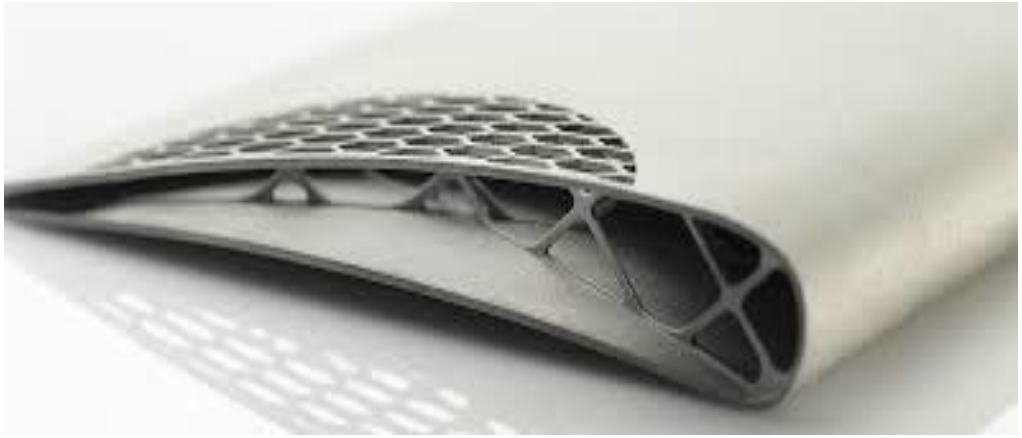
Strong, Lightweight, Customized Parts



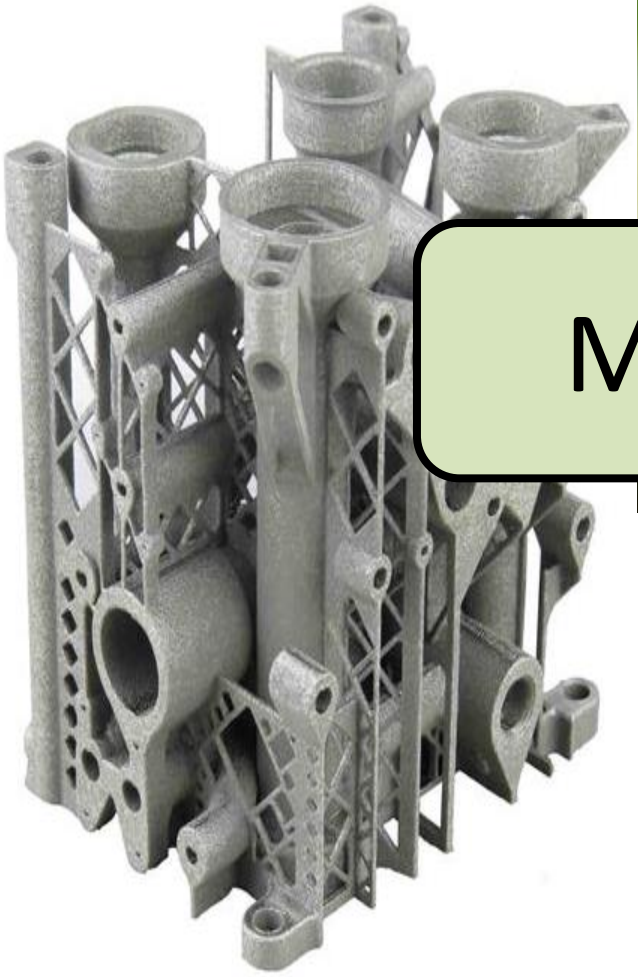








…with Small, Complex Features

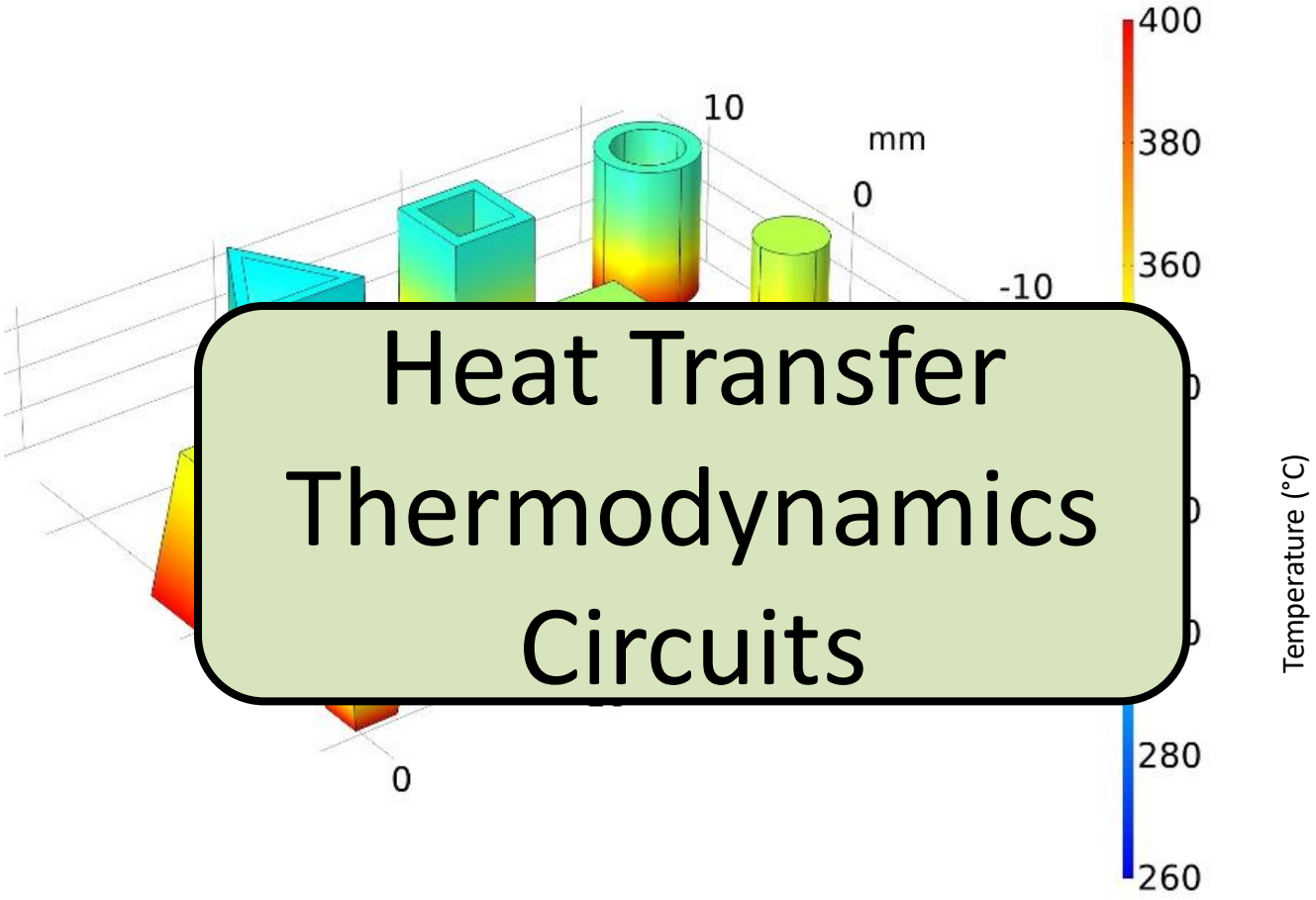




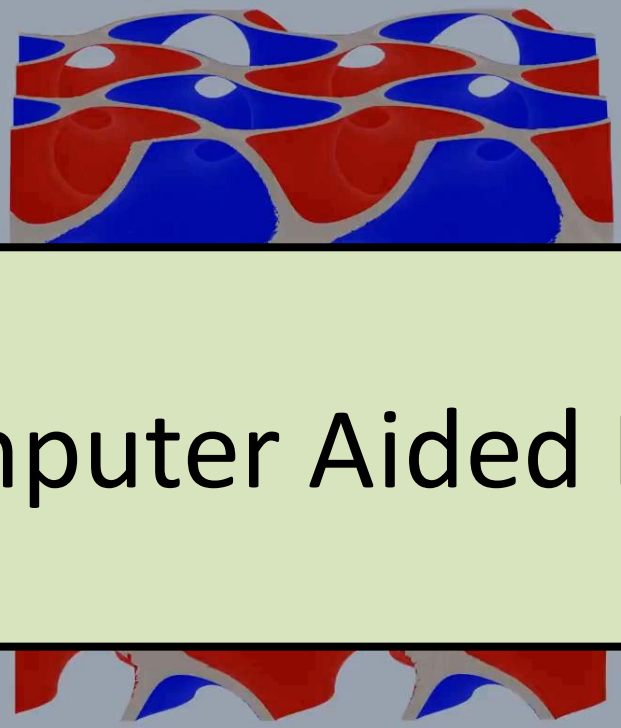
Manufacturing



Predicting Device Performance



Device Design



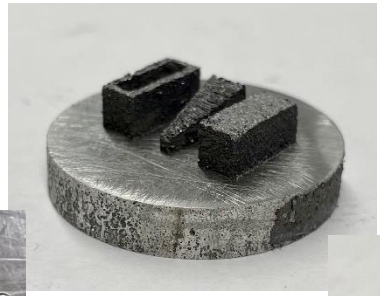
Computer Aided Design

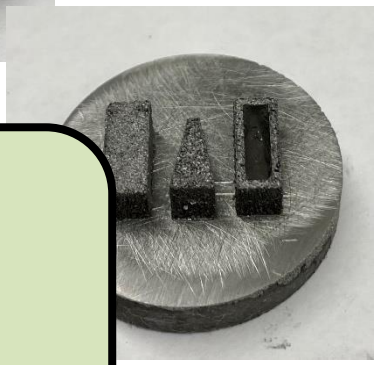
Prototypes

Laser powder bed fusion (selective laser melting):





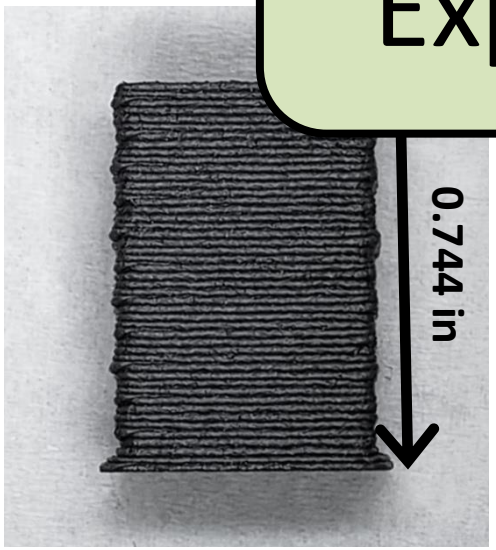


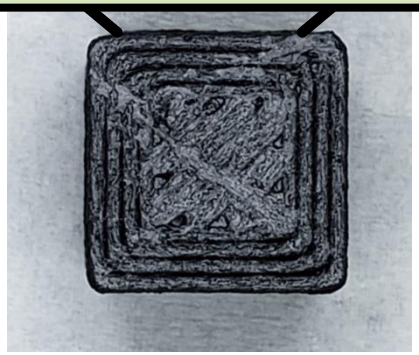


H. Zhang, S. LeB
& Performance

Material ex

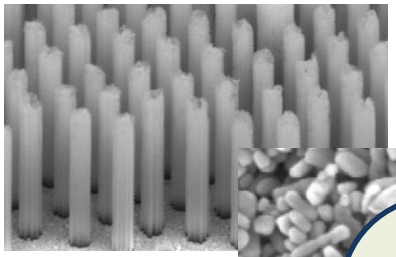
Mechanical Design
Experimental Methods



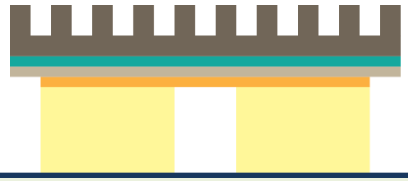


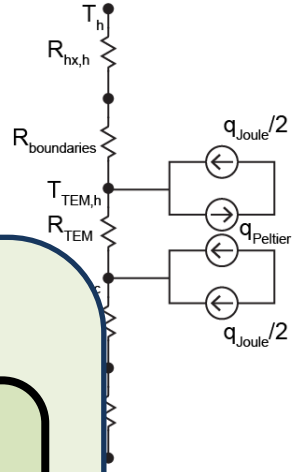


Cost-Performance Metric







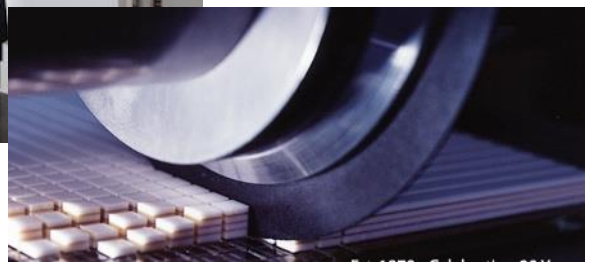


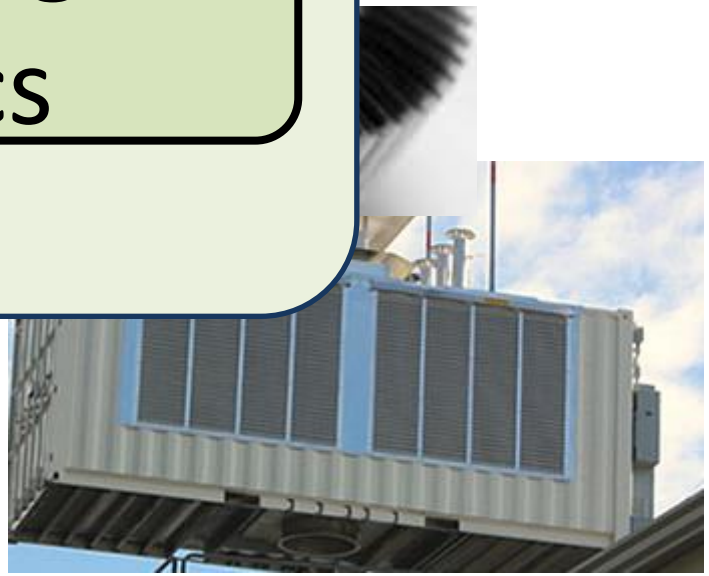
Comprehensive metric combining

Engineering Economics

$/Watt

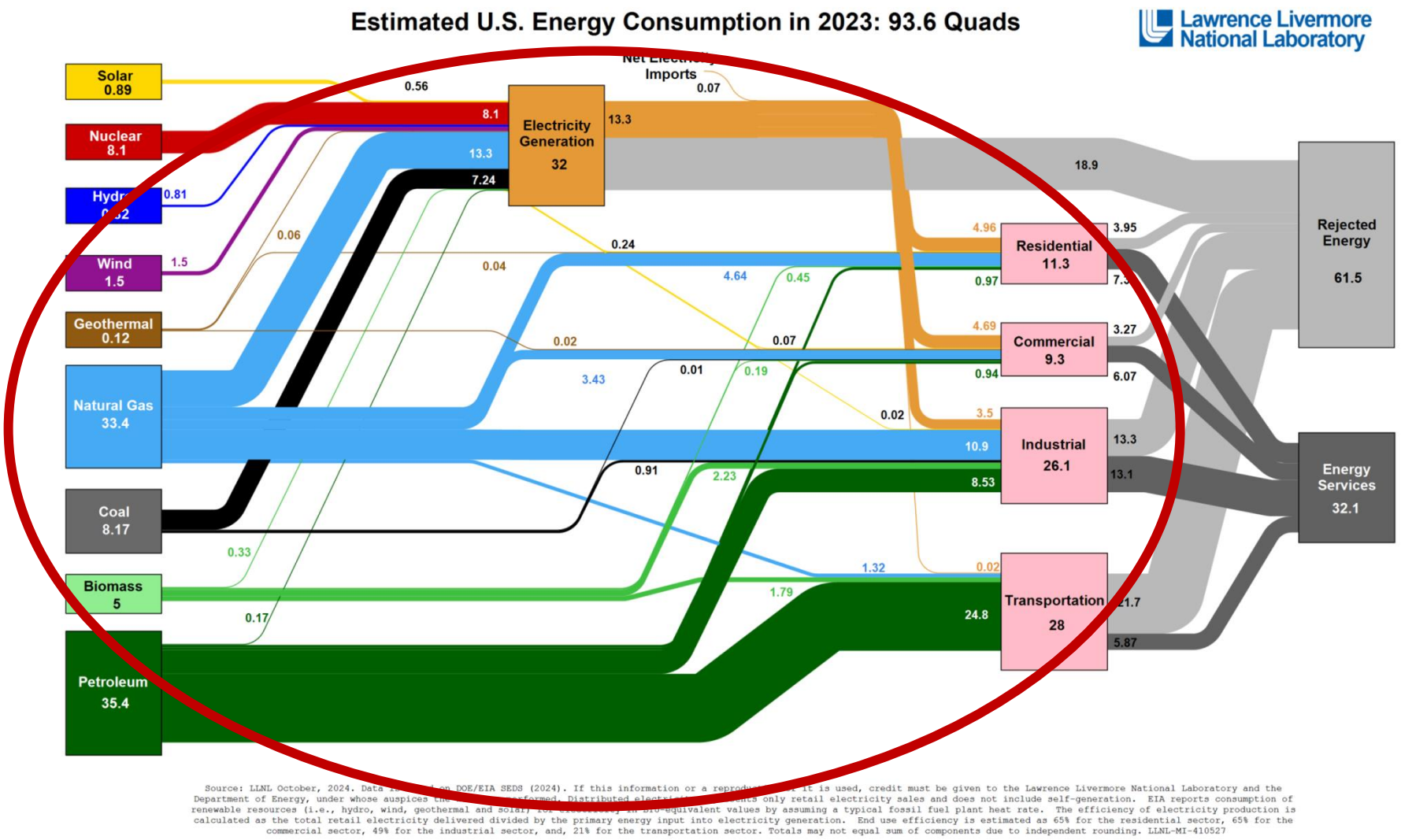






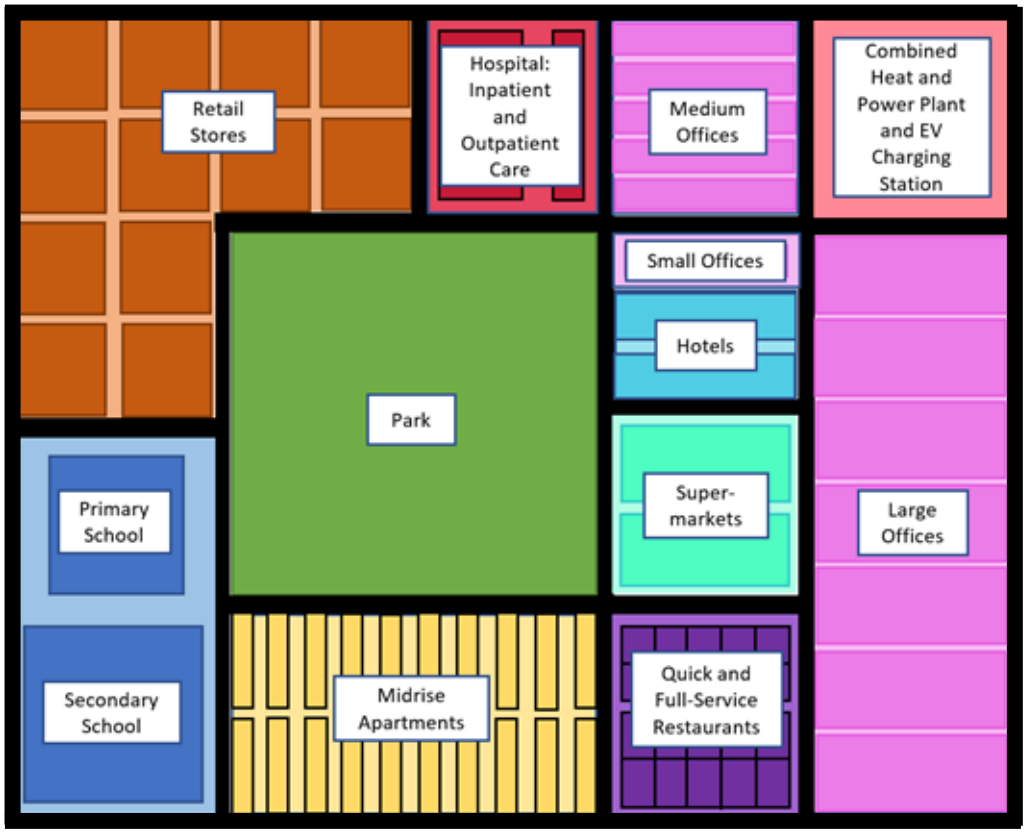
S. LeBlanc *et al., "*Material and Manufacturing Cost Considerations for Thermoelectrics," *Renewable & Sustainable Energy Reviews* (2014)

How do we manage all that energy?



https://flowcharts.llnl.gov/commodities/energy

Example Urban District Energy System



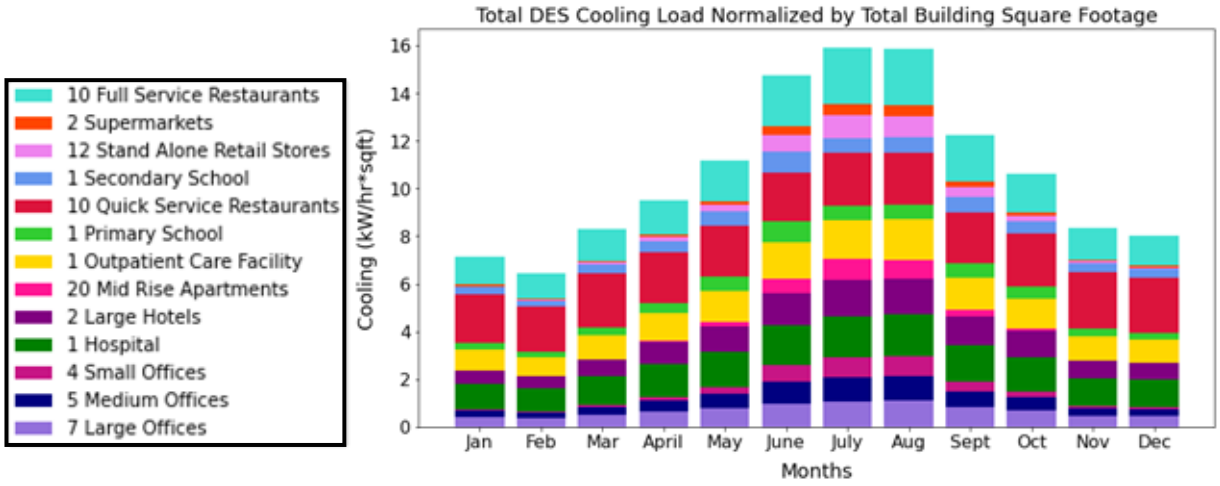


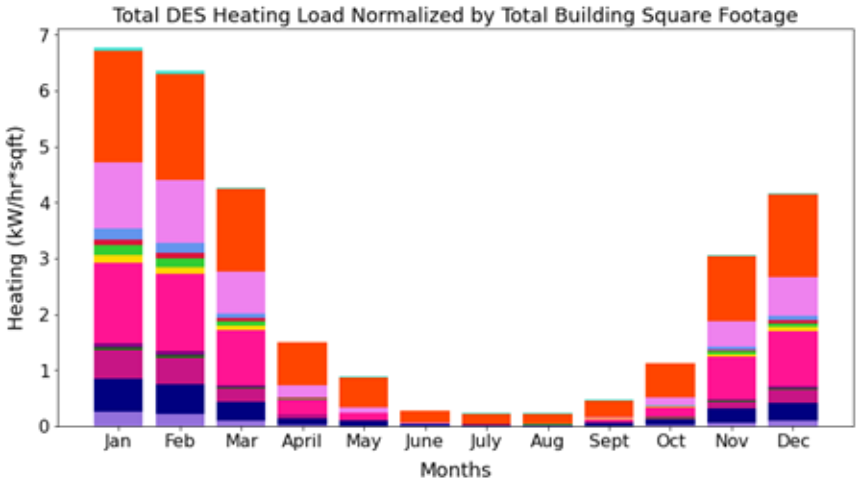
= One City Block (100,000 square feet)

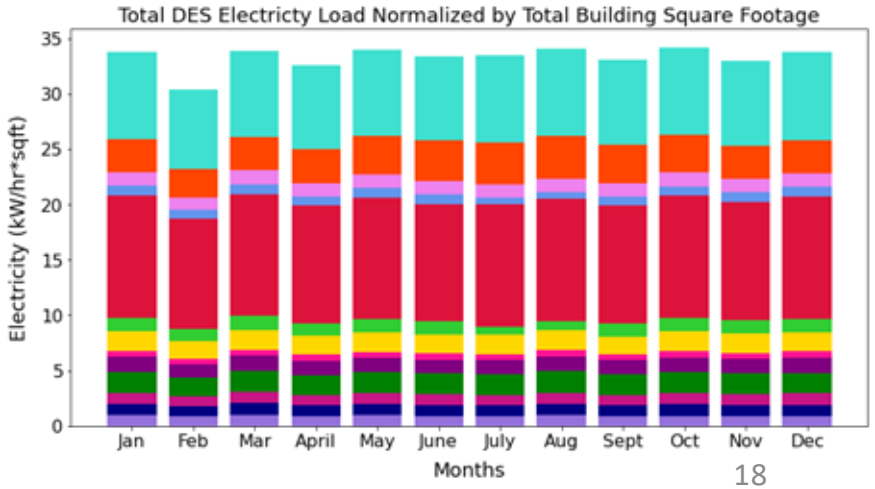
- 1 Primary and 1 Secondary School
- 20 Midrise Apartments
- 10 Quick Service and 10 Full-Service Restaurants
- 7 Large, 5 Medium, and 4 Small Office Buildings
- 2 Supermarkets
- 2 Hotels
- 1 Combined Heat and Power Plant, and 1 EV Charging Station
- 1 Hospital with Inpatient and Outpatient Care
- 12 Retail Stores

- Climate Zone 4A
- Combined Heat and Power Plant
- Hot and Chilled Water Thermal Storage
- Battery Storage
- Solar Photovoltaic Panels
- Electric Vehicle Charging Station

Energy Loads



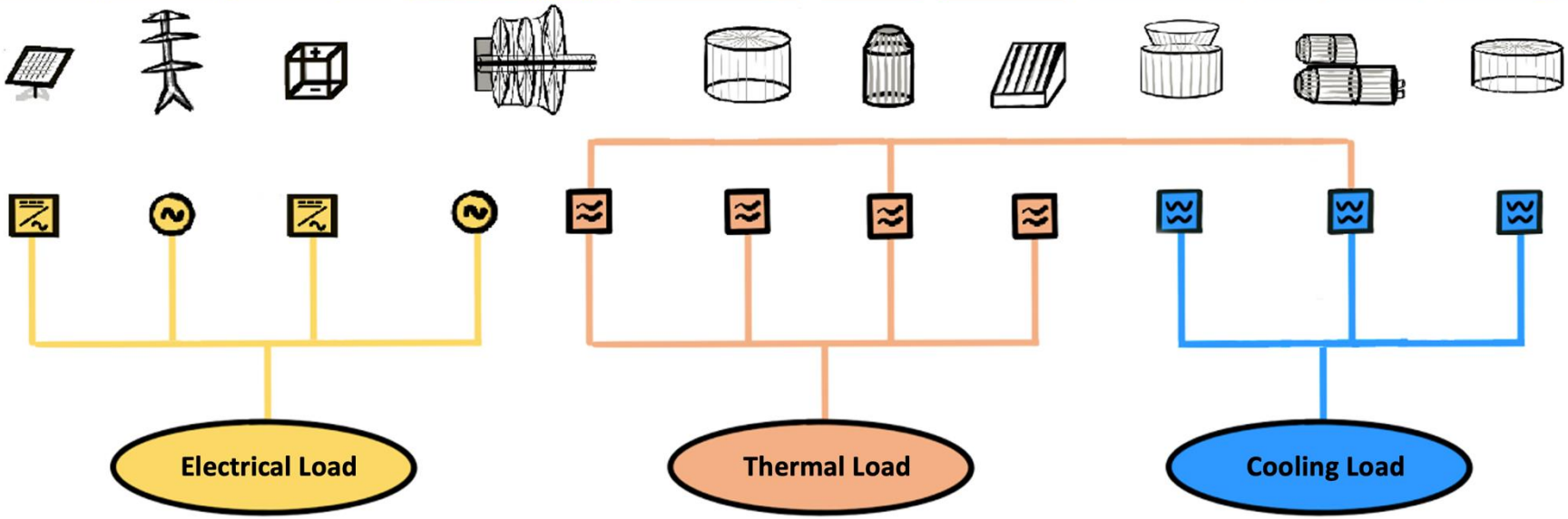




(normalized by building square footage)

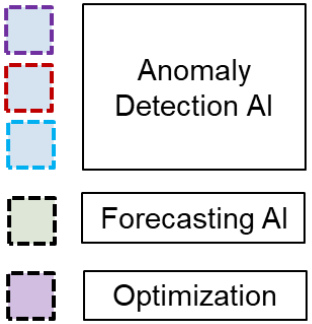
Energy Management System

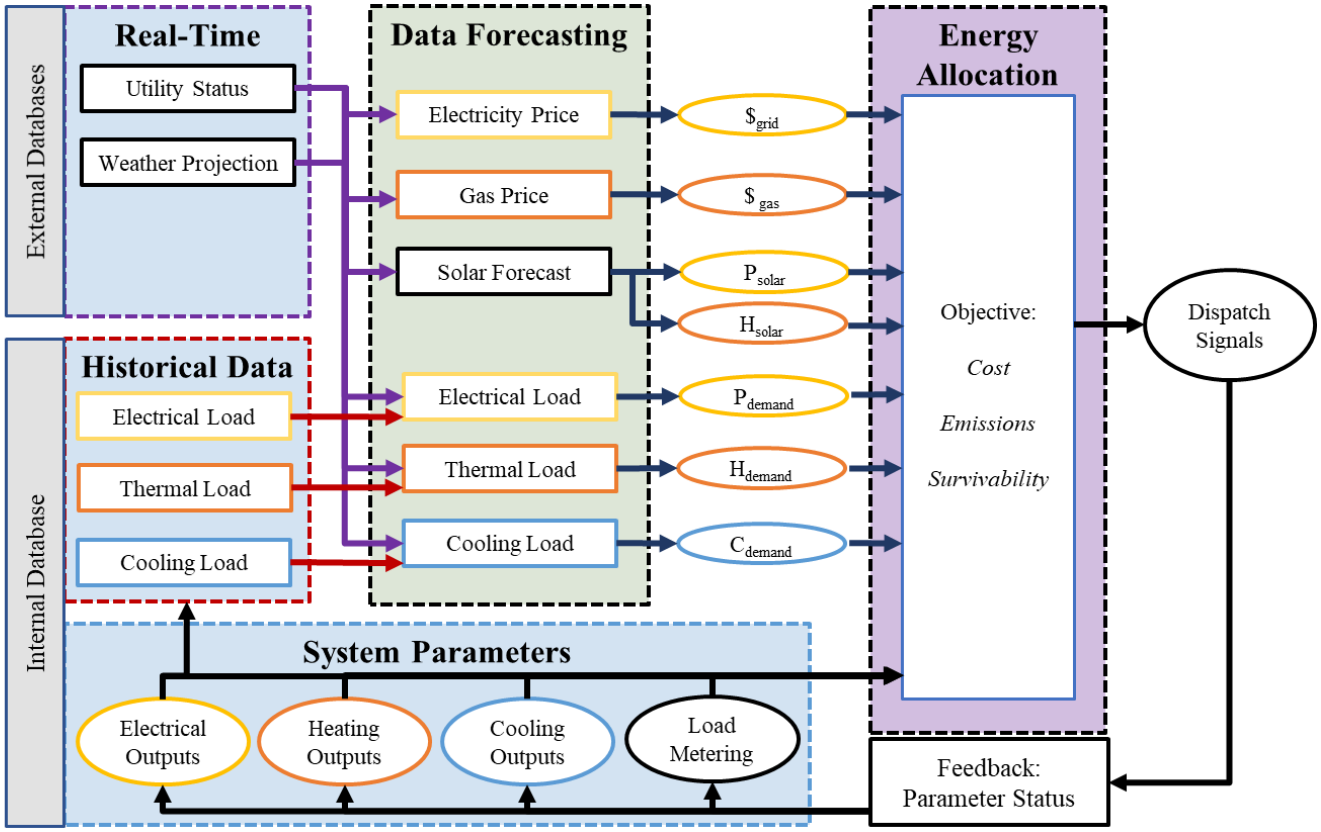
Parameter	Power Output	Max In/Out	Capacity	Heat Output	Capacity	Heat Output	Heat Output	Output Max	Output Max	Capacity
	[kW]	[kW]	[kWh]	[kW]	[kWh]	[kW]	[kW]	[gpm]	[gpm]	[gal]
	5491	5000	24071	1538	19569	23503	0	26024	2246	1754995
Sizing	Area	Min In/Out	Critical Load Support	Heat Load Percentile	Critical Load Support	% Peak Heat	Area	% Peak Cooling	% CHP Peak Heat	Critical Load Support
	[m^2]	[kW]	[days]	[%]	[days]	[%]	[m^2]	[%]	[%]	[days]
	49139	0	0.50	75	0.50	125	0	125	100	0.50
Energy Resource	Solar	Grid	Battery	CHP	Thermal Storage	Boiler	Solar Heater	Towers & Chiller	Absorption Chiller	Ice Storage



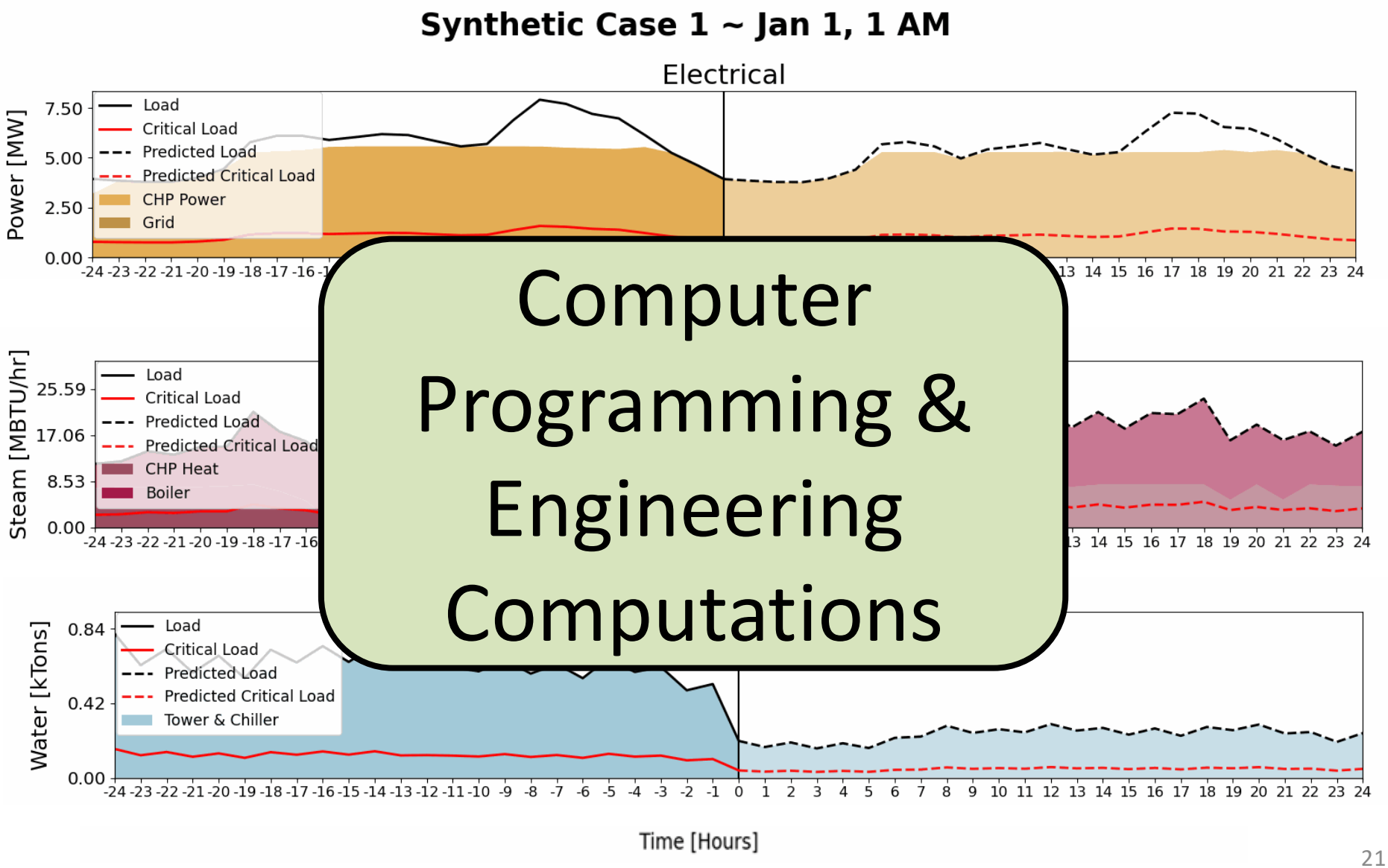
EMS Architecture

- Designed for general DESs
- Utilizes AI & optimization techniques to allocate energy resources



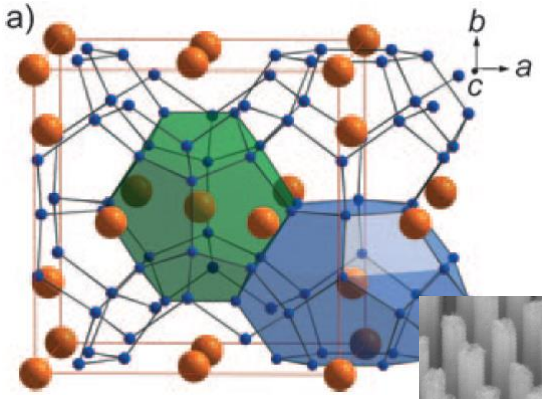


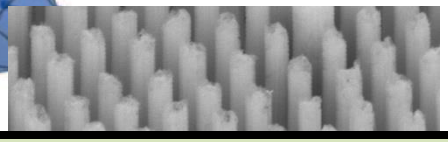
EMS Operation



Computer Programming & Engineering Computations

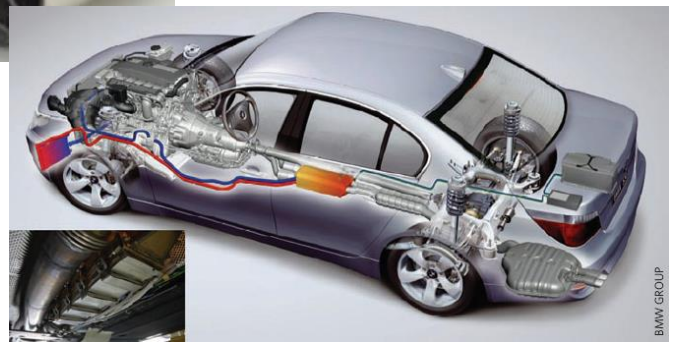
The Many Facets of Energy





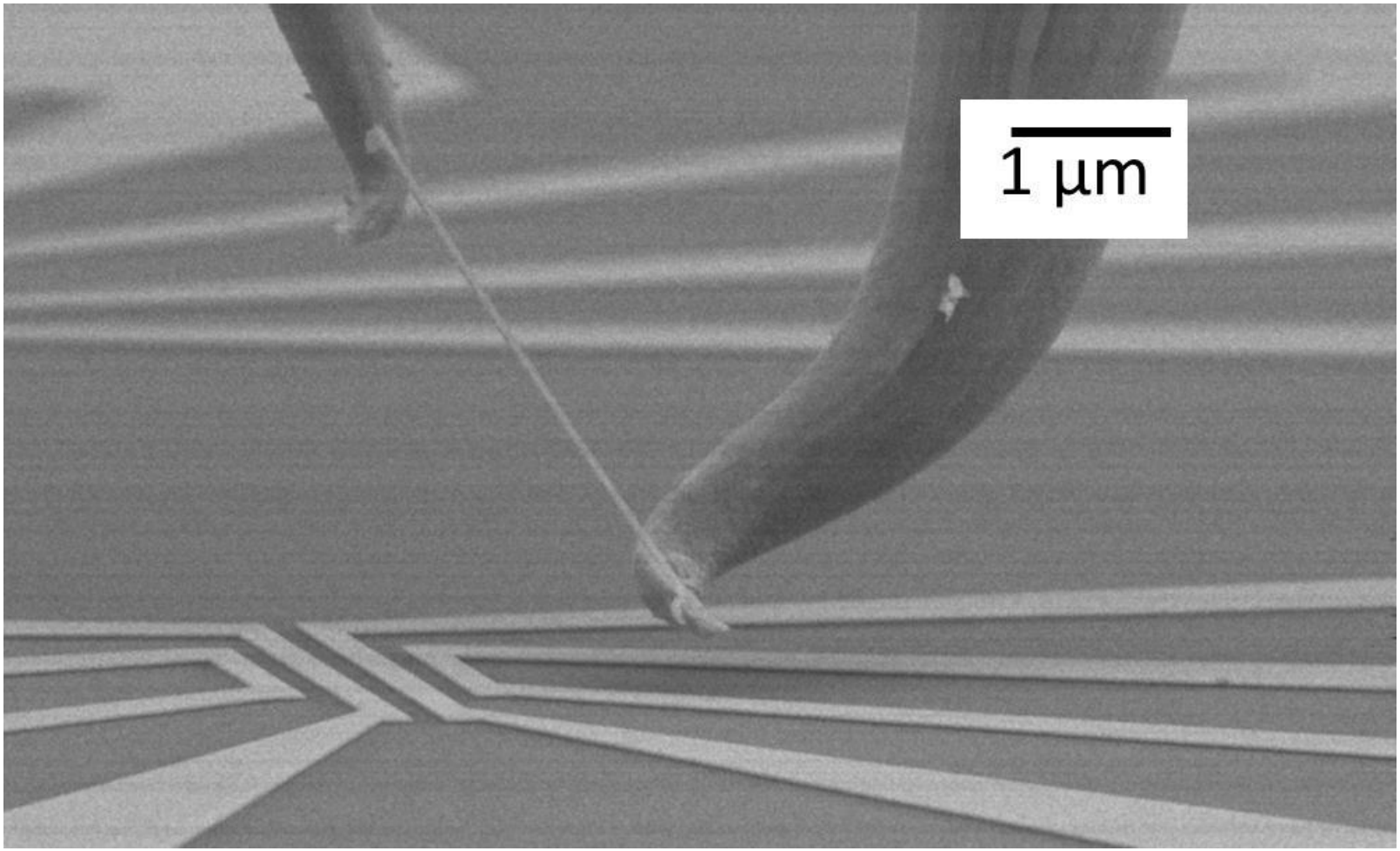
Mechanical & Aerospace Engineering





Questions?

sleblanc@gwu.edu
www.leblanclab.com



S. LeBlanc *et al., J. of Heat Transfer,* Photogallery, August 2012

Supported by DOE Center for Integrated Nanotechnologies